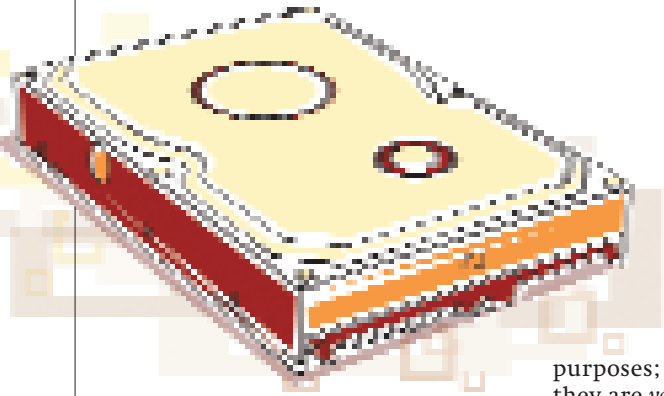




Hard Drives

Hard drive prices seem to have hit an all-time low... or will they drop again?

Storage capacities and speeds have constantly been on the rise the past few years, and older technologies have made way for newer and faster ones. Prices are amazingly low. But even with such a no-frills component, there's a lot you need to look at before buying one.



What You Should Be Looking At

Capacity: Naturally, the most important parameter of a hard drive. Your first thought will be to just choose a capacity sufficient for your perceived needs, and when someone suggests a larger capacity, you tend to ask yourself: "Who needs so much storage space?" It is, however, a real-world observation that it is a good idea to purchase either (a) a disk of a capacity either 1.5 times what you think your requirements are, or (b) the next higher capacity drive from what you're looking at now. Purchasing later is often more expensive. Remember that even as hard drive capacities grow exponentially, so do your needs! The good news is that you can get even big hard disks (read 250 GB and above) at a low price, so if you're looking for a basic home solution, you won't need to skimp on space.

Buffer memory and spindle speeds: The buffer memory affects the performance of a hard drive when there is a lot of data transfer. 8 MB should be the minimum that you should opt for. The higher the spindle speed, the better the performance, but the higher the price. 7,200 rpm is enough for most

purposes; 10,000 is for enthusiasts—they are *very* expensive.

Interface: IDE is fast fading away; newer motherboards are doing away with that legacy interface altogether, sporting a single IDE connector for the optical drive. It therefore makes sense to go with a SATA-II drive, even if an IDE drive is much cheaper.

Analyse Your Usage

General home use: 160 GB is enough for general home use. You can still store lots of movies and music and even games and still have space to spare.

Gaming: Most new games come on dual-layer DVDs and therefore occupy capacities in excess of 10 GB when installed. A 320 GB hard drive based on perpendicular recording technology is therefore advisable—or two smaller, identical drives in a RAID 0 configuration. In the latter case, you get a performance gain, but it's less secure than in the former case, and more expensive as well.

AV professional: Data capacity needs are the highest in this case, more so if the video is in HD. A 1 TB hard drive still costs too much, so it makes sense to purchase two 500 GB disks and create a RAID 0 array. Apart from space, this will mean you get the speed required to capture HD content. If you can afford it, opt for a 15,000 rpm SAS (Serial Attached SCSI) drive. Certain manufacturers have developed an AV class of drives tailored for AV professionals, but these are twice as expensive as regular drives.

Internal / external: An internal drive is the default choice; an external one is useful where data needs to be transferred from one computer to another, or when a system administrator requires to back up data from several computers that are not on a network.

Frills

Some drives come with frills such as cooling fins and clear-glass cover that lets you look at the hard drive in action. While the cooling fins are essential in a high-speed hard drive to dissipate the excessive amount of heat generated, the clear-glass cover is simply a "luxury" feature—cosmetic, if you will.

Innovations

It looks like solid state hard drives will soon phase out the magnetic-media-based drives of today. Samsung has announced the development of the first 64 Gigabyte NAND Flash memory chip using a 30 nm production process, which paves the way for



companies to produce SSD (Solid State Disk) drives with capacities up to half a terabyte. The advantages are the very high speeds as well as extremely low power consumption. There aren't any moving parts, so there's no "spin-up" time (you know the sound—that annoying whine when you start up)—expect your OS to start faster because of this, and even faster thanks to the significantly higher data access speeds. Several vendors have started selling these drives, though they're targeting laptop users more. Unfortunately, you can't get your hands on drives over 160 GB in size—and even if you could, you should know that just the 32 GB drives cost \$350 (Rs 14,000), bringing the cost per GB to a whistle-inducing Rs 437! All we can do is wait, for now... ■



Agent Tips

As of now, 320 GB is the sweet spot for internal hard drives in terms of price. A drive with 8 MB of buffer memory costs around Rs 3,200, while one with a 16 MB buffer costs about Rs 4,000.